

Difference-Integrated Fixed-Point Diffusion: A Structural-Dynamical Hybrid Framework and Its Application to Cross-Species Body-Language Interfaces

Masaomi Chiba

November 29, 2025

Abstract

We propose *Difference-Integrated Fixed-Point Diffusion* (DIFD), a general-purpose framework combining isotropic diffusion processes with a non-equilibrium structural operator derived from the Derridean notion of *différance*. The DIFD update rule integrates a symmetric noise term with an asymmetric, history-sensitive structural gradient. This hybrid mechanism allows an agent or model to search not only within a likelihood-driven manifold but also across structural variations that are typically ignored by purely statistical samplers.

A formal definition of orthogonality between diffusion noise and structural gradient is provided, along with empirical validation protocols. As a concrete application, we demonstrate how DIFD enables the construction of a “cross-species body-language interface”—a tool that infers and synthesizes body-language cues enabling humans to communicate more effectively with domestic animals, focusing on cats as a primary example.

1 Introduction

Diffusion models have become a core generative paradigm, typically exploring a data manifold through noise-perturbed score matching. However, these models remain restricted by the statistical structure of the data distribution. They lack a mechanism to traverse *structural* alternatives—variations in meaning, relationality, or tacit behaviour that are not simply captured by likelihood gradients.

To address this, we introduce *Difference-Integrated Fixed-Point Diffusion* (DIFD), which augments diffusion dynamics with a structural correction term inspired by Derrida’s *différance*: the notion that meaning arises from both difference and deferred relation across representations. The structural component is formalized as a non-symmetric “difference operator” acting as a gradient of a cost functional C defined over relational structures rather than mere pointwise data.

One surprising consequence is that the hybrid sampler, when properly tuned, can be used to synthesize relational cues for embodied systems. This enables applications such as cross-species body-language mediation for animal behaviour support.

2 Difference-Integrated Fixed-Point Diffusion

2.1 Update rule

Let $z_t \in \mathcal{Z}$ be the latent state at time t . DIFD applies the update

$$z_{t+1} = z_t + \epsilon_t - \lambda_t \nabla_z C(z_t), \quad (1)$$

where $\epsilon_t \sim \mathcal{N}(0, \sigma_t^2 I)$ is isotropic diffusion noise and C is a structural cost functional.

The cost C can encode:

- relational asymmetries,
- deferred dependencies (temporal or multi-scale),
- behavioural counterfactuals,
- morphological constraints.

Thus ∇C introduces structural bias rather than statistical bias.

2.2 Interpretation as hybrid search

Diffusion drives isotropic local exploration; the DIFW operator drives directed structural movement. When orthogonality (defined below) approximately holds, the components decouple, enabling efficient structural traversal without corrupting stability.

3 Formal Orthogonality

3.1 Definitions

Let $g_t := \nabla_z C(z_t)$.

Second-moment orthogonality.

$$\text{Cov}(\epsilon_t, g_t) = \mathbb{E}[\epsilon_t g_t^\top] = 0. \quad (2)$$

Information-theoretic orthogonality.

$$I(\epsilon_t; g_t) \approx 0.$$

3.2 Why orthogonality matters

Consider:

$$\mathbb{E}\|z_{t+1} - z_t\|^2 = \mathbb{E}\|\epsilon_t\|^2 + \lambda_t^2 \mathbb{E}\|g_t\|^2 - 2\lambda_t \text{Tr}(\mathbb{E}[\epsilon_t g_t^\top]).$$

Under orthogonality the cross-term vanishes.

3.3 Geometric sufficient condition

Let $\Sigma_t = \text{Cov}(\epsilon_t)$. If g_t lies largely in the subspace corresponding to small eigenvalues of Σ_t , the covariance vanishes. With isotropic noise this reduces to the requirement that g_t has zero expected correlation with ϵ_t .

3.4 Empirical validation

We recommend the following tests:

1. empirical cross-covariance test,
2. directional correlation profile against principal axes of ϵ_t ,
3. mutual information estimation (e.g. k-NN estimator),
4. intervention: enforce alignment between g_t and noise directions and measure degradation.

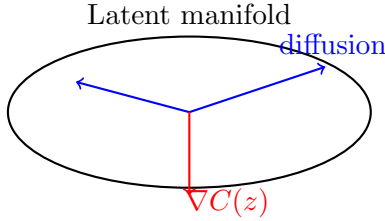


Figure 1: Diffusion directions and DIFW gradient approximately orthogonal.

3.5 Illustration

4 Application: Cross-Species Body-Language Interface for Cats

The core insight is that DIFD separates:

- *likelihood-driven* motion patterns learned from data,
- *structural body-language operators* capturing deferred relational states.

This makes it suitable for a behavioural mediator bridging species with distinct semiotic systems.

4.1 Modelling feline body-language structures

Cat body language involves:

- tail orientation,
- ear rotation,
- paw tension,
- dorsal curvature,
- micro-movements of whiskers.

These variables are **not** independent. For example:

$$\text{ear angle} \rightarrow \text{tail motion} \rightarrow \text{pupil dilation}$$

forms a deferred dependency chain. Let $C(z)$ encode the mismatch between these relational constraints and candidate generative outputs.

4.2 Body-language synthesis

Given a human cue h , we sample a feline-compatible pose z from:

$$z = \text{DIFD}(h).$$

Examples:

- generating a non-threatening crouch posture,
- indicating invitation-to-approach with slow-tail-swivel,
- conveying playfulness without predatory misinterpretation.

4.3 Behavioural mediation tool

The final tool consists of:

1. a vision module capturing cat posture,
2. a latent translator mapping to relational state,
3. a DIFD sampler generating compatible human-body cues,
4. a feedback loop for continuous adjustment.

The structural operator prevents the system from producing statistically plausible poses that nevertheless violate the tacit relational grammar of feline behaviour.

5 Conclusion

We introduced DIFD, a framework integrating symmetric diffusion with asymmetric relational structure. A formal orthogonality assumption is provided with validation protocols. Finally, we demonstrated how DIFD naturally supports generative relational mediation tasks, including a cross-species body-language interface enabling safer, more intuitive interaction with cats.

References

- [1] Song, Yang, et al. *Score-Based Generative Modeling*. (2021)
- [2] Sohl-Dickstein, Jascha, et al. *Deep Unsupervised Learning using Nonequilibrium Thermodynamics*. (2015)